ALICE L WALTON SCHOOL OF MEDICINE

β – Agonists

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Adapted from: Sympathomimetic Agents

Listen to this Brick
0:00 / 18:13 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Classify beta adrenergic agonists based on their receptor selectivity.
2. Predict the effects of beta-adrenergic agonists considering their receptor selectivity on various organ systems.
3. Explain the mechanism of action and provide a rationale for the clinical indications of beta-adrenergic agonists.
4. Explain the ~~potential adverse effects of beta-adrenergic~~ agonists and discuss strategies to minimize/prevent such effects.

CASE CONNECTION

LK, 14-year-old was warming up for a high school soccer game at the home field. It was a 30-minute bus ride on a very hot and humid day. The windows on the bus were opened to make the team comfortable. The wind was blowing as the team got off the bus. He completed his warm-up rounds, but soon after, he started feeling tightness around his chest. He knew the feeling since he was diagnosed with asthma. He walked to the sideline, grabbed a drink, and started wheezing. He started rummaging through his gym bag for his rescue inhale and panicked when he couldn't find it. LK's breathing was getting harder. No matter how many times his asthma had flared up, he never got used to the feeling of not being able to breathe. He got his mom's attention in the stands, and they drove quickly to the closest emergency room.

What is in the rescue inhaler that can help LK? How do these rescue inhalers work?

LK also asks his mother why he has to use an inhaler and why can't he take a tablet every day to prevent this reaction.

Consider your answer as you read, and we'll revisit LK at the end of the brick.

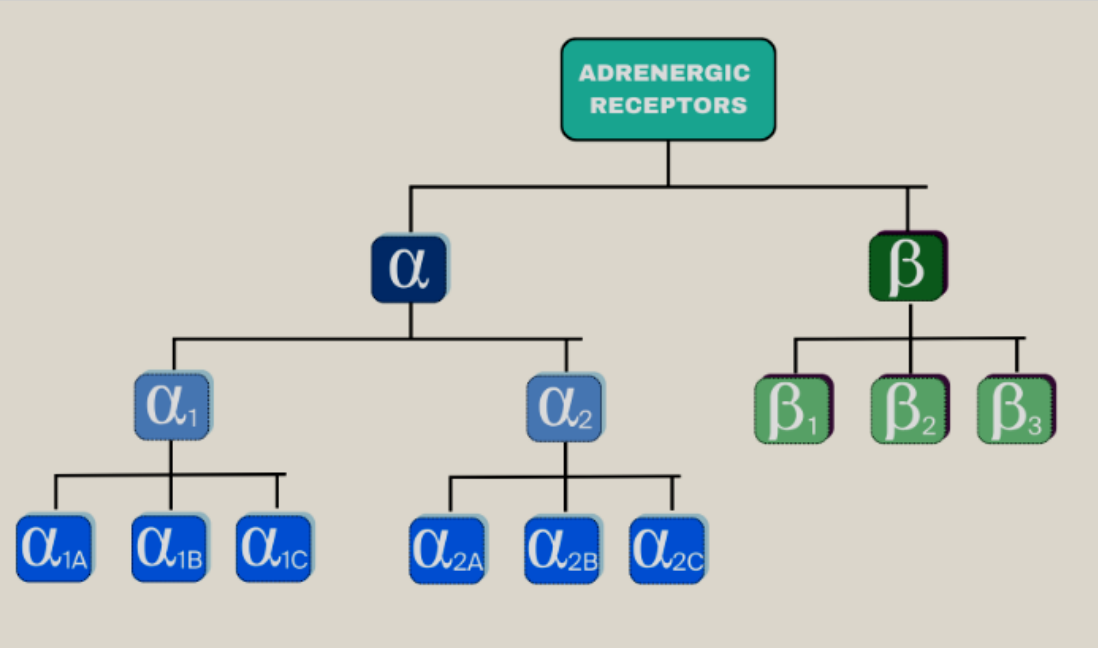
GO TO CONCLUSION ↓

What Are Sympathomimetic agents?

In the previous bricks on the Autonomic Nervous System, we discussed how sympathomimetic drugs stimulate the α-adrenergic and β-adrenergic autonomic receptors, mimicking (partially or completely) the actions of norepinephrine or epinephrine (endogenous neurotransmitters). This brick discusses drugs that are agonists of the β-receptors. Most of the agents are direct-acting β-agonists. This would be a good time to review physiological functions of the β-receptors in various organs in the ANS brick.

How are β–agonists classified? (LO#1)

β-agonists are classified based on their receptor selectivity. If you recall from the ANS brick, β-agonists are classified into β1, β2, and β3.



These are GPCRs (G-protein coupled receptors) that effect downstream signaling, mediating various physiological functions. While β-receptors are expressed in various organs we will focus on some of β-agonist action in the heart and lungs.

Activation of β-1 receptors in the heart causes

ACTIVE LEARNING QUESTION

Review: Match the appropriate GPCR receptor to its downstream signaling.

Gi	decrease in PKA activation
Gs	✓ increase cAMP levels --> increase in PKA -->
Gq	✓ increase IP3--> increase in cytosolic Ca

Explanation

G_s activation increases adenylyl cyclase activation and increases cAMP levels thereby causes activation of downstream activation of PKA. In the heart, this can lead to increase in Ca^{2+} hence increasing contraction and heart rate. G_i is the opposite action of G_s, thus decreasing cAMP levels and downstream activation of signaling processes. G_q activation increases Ca levels, which can lead to muscle and/or vasculature contraction.

What are effects and indications for β–agonists based on their selectivity? (LO#2 and 3)

To predict the effects of β-agonists, it is important to consider the organ and the β receptor isoform selectivity of the agonist. Let's first begin with β1-receptor agonists.

β_1–Agonists: Inotr

One of the organs where β-1 receptors are predominately expressed is the heart. Two agents with largely β_1 activity are dobutamine and isoproterenol.

Dobutamine is an inotrope used to increase cardiac contractility. Dobutamine predominant β_1 agonist action can increases the heart rate (chronotropy) and contractility (inotropy) ($\beta_1 >> \beta_2$). Hence, it is a positive inotropic and chronotropic agent (Recall: β_1: Gs--> increases PKA-->Increasing Ca--> causing contraction). Given these actions, dobutamine is clinically used in cases where an increase in cardiac contractility is need. These would include acute conditions such as heart failure or cardiogenic shock. In these acute care disease states, the heart fails to function and hence an inotropic and/or chronotropic agent is required to initiate contraction and restore circulation.

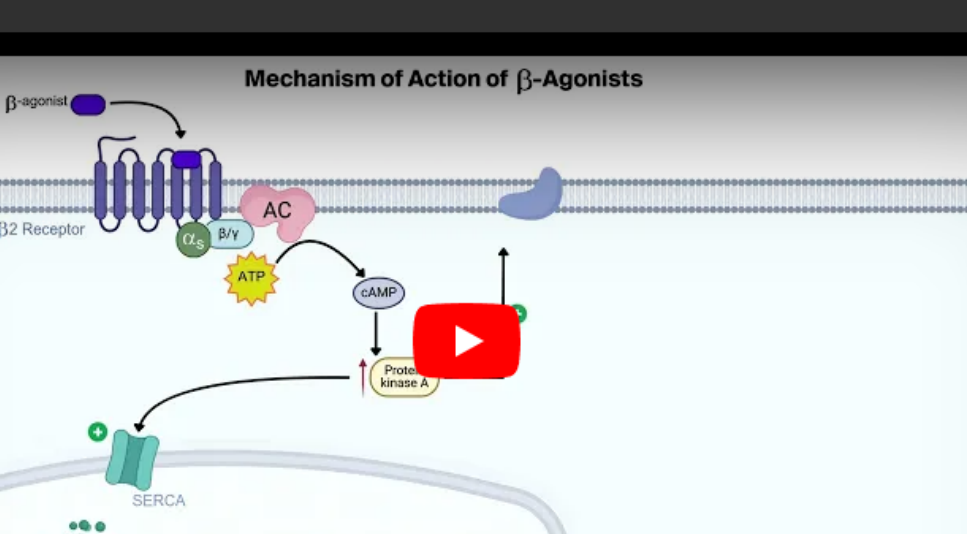
Isoproterenol (or isoprenaline) has stronger β_2-agonist effects than β_1 ($\beta_2 >> \beta_1$). The β_2 activity of Isoproterenol causes more bronchodilation and vasodilation. Thus, it is used in acute treatment of asthma and COPD (although rarely). However, there are far better β_2 selective agents that are available for the treatment of asthma. The β_1 activity may be helpful in treating bradycardia and heart block. This use is very limited, and we will discuss it further in the cardiovascular organ system.

What is the primary action of Dobutamine?

β_2–Agonists: Bronchodilation

The β_2 receptors are expressed in the lungs and hence β_2 agonists are largely **used as bronchodilators**, since β_2-receptors are dense in the bronchioles and cause relaxation of the smooth muscle there.

β_2-agonists relax the bronchial smooth muscles. The video below describes how this bronchodilation occurs.


Mechanism of Action of β-Agonists
β-agonist
β2 Receptor
AC
αs
β/γ
ATP
cAMP
Protein kinase A
SERCA
ER (Ca store)
smooth muscle cell cytoplasm
Created in https://BioRender.com

AC= Adenlyl Cyclase, SERCA= Sarcoplasmic/Endoplasmic Reticulum Ca-ATPase Pump

Commonly used β_2-agonists include **Albuterol, Salmeterol, Formoterol,** and **Terbutaline ("-terol" suffix except for Terbutaline).** Since they relax bronchial smooth muscles, β_2-agonists are commonly used in respiratory disorders such as asthma and Chronic Obstructive Pulmonary Disorder (COPD) (Figure 2). Depending on the chemical structure of the β_2-agonist, they are classified as short-acting β_2-agonists (SABA) or long-acting β_2-agonists (LAMA). Albuterol is an example of SABA, while Salmeterol, Formoterol are examples of LABA. Given that albuterol is a SABA, they are used as a 'rescue' inhaler for Asthma and COPD patients. LABA's, on the other hand are used when the disease worsens or begins to become resistant to treatment.

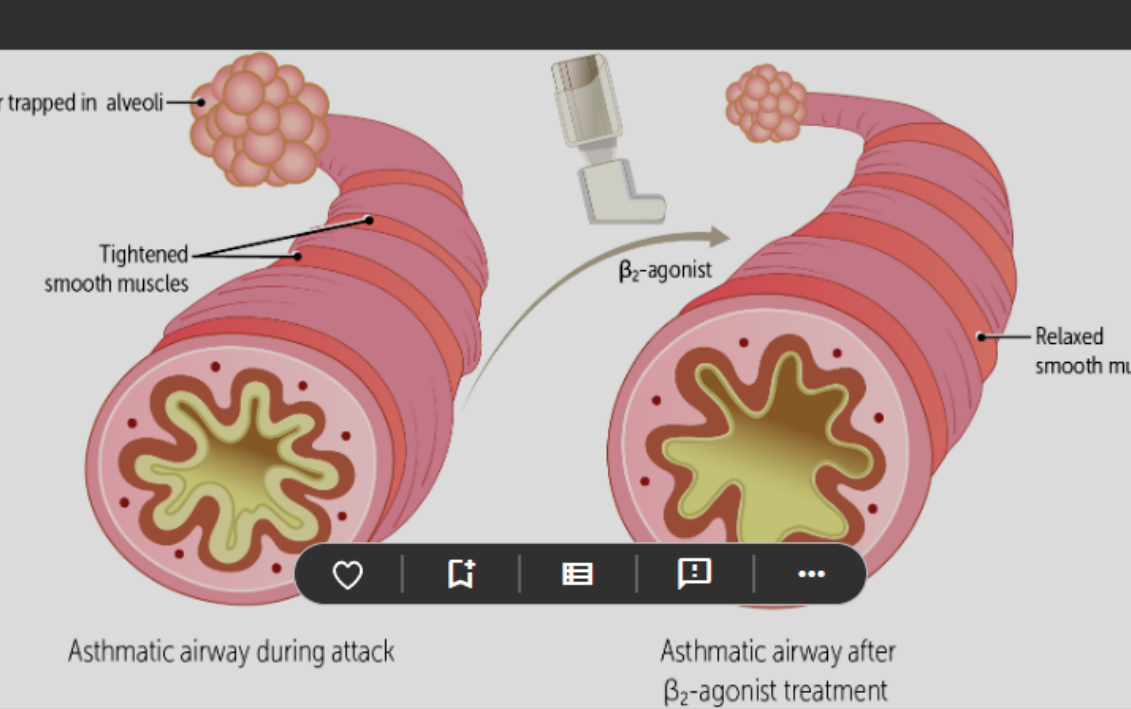


Figure 2

The inhaler route of administration provides an easy, convenient way to administer with rapid onset of action. In addition, it decreases the risk of systemic side effects (in comparison to oral administration).

β_2-agonists can activate the Na^+/K^+-ATPase pump. The Na^+/K^+-ATPase pump functions to drive Na out of the cell and K into the cell. This helps to shift potassium into the cell. Pushing the K into the cell (or shifting) can help remove excess K from the serum. Based on this mechanism, β_2-agonists (typically Albuterol) can be used in the **treatment of hyperkalemia.** Terbutaline is used off-label as **tocolytic agents** in preterm labor because they relax the uterine smooth muscle.

β_3–Agonists: Detrusor smooth muscle relaxation

β_3 adrenergic receptors play a key role in regulating detrusor smooth muscle tone and enhancing the bladder's storage function. **Mirabegron**, a β_3 agonist, works by activating β_3 receptors in the bladder. This activation relaxes the detrusor smooth muscle during the urine storage phase, thereby increasing bladder capacity. Mirabegron is considered selective for the $\beta_3 > \beta_2$ and β_1.

What are the main clinical uses of β_2-agonists?

What is the primary clinical use of Mirabegron?

What adverse e β–agonist use? (LO#4)

The ADRs of the agents depend on the receptor subtype and the indicated use of the drug.

ADRs for β1-agonists

- Dobutamine use is most commonly associated with tachycardia or tachyarrhythmia (arrhythmia associated with increased heart rate). This arises from β_1-agonist action.
- Dobutamine can also cause peripheral extravasation.

What is a common adverse effect of dobutamine?

CLINICAL CORRELATION

When vasopressors , extravasation occurs. This can cause severe local vasoconstriction, potentially leading to skin necrosis. To prevent this, it's recommended to administer vasopressors through a central vein whenever possible. If infiltration does occur, injecting phentolamine (an α-antagonist) subcutaneously can help reduce local vasoconstriction.

ADRs for β_2-agonists

- Muscle Tremors: Stimulation of β_2 receptors in the skeletal muscle can cause tremors. This is a very common effect from β_2-agonist use.
- Tachycardia and palpitations can also occur. This is due to the atrial activation of β_2-receptors.
- Hypokalemia can potentially occur due to the K shift. Although profound hypokalemia is rare, β_2-agonists reliably reduce potassium levels.

What are common ADRs of β2-agonists?

ADRs for β_3-agonists

Urinary retention: Given its β_3 properties urinary retention is a common adverse effect. Hypertension is also a common adverse effect. In fact, patients with severe or uncontrolled hypertension should not be taking Mirabegron.

What are common adverse effects of β3-agonists like Mirabegron?

CASE CONNECTION

BACK TO INTRODUCTION ↑

What is in the rescue inhaler that can help LK? How do these rescue inhalers work? LK also asks his mother why he has to use an inhaler and why can't he take a tablet every day to prevent this reaction.

When LK reached th the rescue inhaler. Albuterol is a β2-ago muscles and decrease the "tightness in the chest" feeling that LK was experiencing. β2 receptors in the bronchial smooth muscles are Gs-type receptors that decrease the cytosolic Ca^{2+}. The β2-agonists are administered as inhalers as they can immediately reach the site of drug action (the bronchial smooth muscles) and illicit drug action. Administration of an oral drug form would delay the onset of action and decrease the bioavailability of the drug (first-pass effect).

Summary

- Sympathomimetic drugs stimulate the α-adrenergic and β-adrenergic autonomic receptors, mimicking the actions of norepinephrine or epinephrine.
- β-agonists are classified based on their receptor selectivity, with β1, β2, and β3 affecting different organs and physiological functions.
- β1-agonists like dobutamine increase heart rate and contractility, and are used in cases of heart failure or cardiogenic shock.
- β2-receptor activation by the β2-agonists increase the uptake of Ca in the bronchial smooth muscles and hence cause bronchodilation.
- β2-agonists, such as Albuterol and Salmeterol, are used as bronchodilators in respiratory disorders like asthma and COPD.
- β3-agonists like Mirabegron regulate detrusor smooth muscle tone, enhancing the bladder's storage function.
- Adverse effects of β-agonists depend on the receptor subtype and use of the drug, and can include tachycardia, muscle tremors, urinary retention, and hypertension.

Review Questions

1. Which of the following medications would be best for use in preventing premature labor?

A. ○ Epinephrine
B. ○ Isoproterenol
C. ○ Phenylephrine
D. ● Terbutaline

Hide Explanation

The correct answer is terbutaline (D). Terbutaline is a β_2-agonist that causes smooth muscle relaxation. β_2-Receptors are found on the smooth muscle surrounding the uterus; β_2-agonists can relax the uterus and help minimize contractions, thus helping to prevent premature labor. This effect is also known as tocolysis. Epinephrine (A) is incorrect because even though it has β_2-agonist effects, it also affects α_1-, α_2-, and β_1-receptors, making it unsuitable for tocolysis. Isoproterenol (B) is incorrect because it has β_1 and β_2 activity, which makes it a less-than-optimal selective β_2-agonist choice for tocolysis. Phenylephrine (C) is incorrect because it is an α_1-agonist, which causes smooth muscle contraction, not relaxation.

2. Which drug is prin

A. ○ Albuterol
B. ● Dobutamine
C. ○ Isoproterenol
D. ○ Terbutaline

Hide Explanation

The correct answer is Dobutamine (B). Dobutamine is primarily a β1 agonist, which increases heart rate and contractility. Isoproterenol (C) has stronger β2 effects than β1. Albuterol (A) and Terbutaline (D) are β2 agonists.

3. Which is a common adverse effect of β_2-agonists?

A. ○ Hyperkalemia
B. ○ Bradycardia
C. ○ Hypotension
D. ● Muscle Tremors

Hide Explanation

The correct answer is Muscle tremors (D). Muscle tremors (D) are common due to stimulation of β2 receptors in skeletal muscle. Hypokalemia can potentially occur due to the K shift, and not hyperkalemia (A). The use of β2-agonists causes tachycardia, not bradycardia (B). Hypotension (C) is not typically associated with β2-agonists.

4. Which of the following is a β_3 agonist?

A. ○ Albuterol
B. ○ Dobutamine
C. ○ Isoproterenol
D. ● Mirabegron

Hide Explanation

The correct answer is Mirabegron (D). Mirabegron is a β_3 agonist that helps relax the uterine smooth muscles. Albuterol (A) is a β_2 agonist used as a bronchodilator. Isoproterenol (C) is a β_2 and β_2 agonist, while Dobutamine (B) is a β_2 agonist.

How would you rate this Brick?

the uterine smooth muscles. Albuterol (A) is a β_2 agonist used as a bronchodilator. Isoproterenol (C) is a β_2 and β_2 agonist, while Dobutamine (B) is a β_2 agonist.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Classify beta adrenergic agonists based on their receptor selectivity.~~
02 ☑ ~~Predict the effects of beta-adrenergic agonists considering their receptor selectivity on various organ systems.~~
03 ☑ ~~Explain the mechanism of action and provide a rationale for the clinical indications of beta-adrenergic agonists.~~
04 ☑ ~~Explain the potential adverse effects of beta-adrenergic agonists and discuss strategies to minimize/prevent such effects.~~

Objective Confidence: 4/4